

**Sexual Communication as a Moderator of the Association Between
Erection Hardness and Sexual Satisfaction Among Male Military
Veterans**

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Proposal Declaration

I, Anfisa Rekina, declare that this proposal has been composed solely by myself and that it has not been submitted, in whole or in part, in any previous application for a degree. Except where stated otherwise, by reference or acknowledgement, the work presented is entirely my own and has been generated by me as the result of my own original research.

I have used AI (ChatGPT) for discussing the structure and coherence of the proposal, refining academic language, and checking reference formatting and bibliographic details, ensuring that all work is in my own words and based on my own understanding.

.....

Signed: Anfisa Rekina

Date: 14.09.2026

Introduction

Sexual Script Theory proposes that understandings of sex are organised through cultural, interpersonal, and intrapsychic scripts (Gagnon & Simon, 1973; Simon & Gagnon, 1986). Cultural scripts provide socially available narratives concerning legitimate sexual behaviours and roles. At the interpersonal level, individuals enact and revise sexual scripts in relation to their identities and experiences, while intrapsychic scripts encompass individual desires and meanings (Pham, 2016). Sexual experiences may consequently be evaluated partly against learned expectations of what satisfying sex should involve.

Empirical research suggests that culturally dominant scripts privilege penile–vaginal intercourse over other sexual practices. Interview studies in the US and France have found persistent heteronormative definitions of sex centred on phallic or penile–vaginal penetration (Pham, 2016; Thomé, 2025), while a larger mixed-methods study (N = 907) similarly identified strong gendered and heteronormative scripts structuring sexual interactions (Harvey et al., 2023). Importantly, penetration-centred scripts carry gendered expectations of sexual performance. For instance, men may be expected to achieve and maintain an erection, perform penetration, and facilitate their partner’s orgasm, allowing sexual performance to function as a marker of masculine competence (Thomé, 2025). Erectile functioning (EF) may therefore acquire particular significance within culturally available understandings of successful sex.

Nevertheless, when EF becomes impaired, penetration-centred expectations of successful sexual performance may become difficult to enact. Erectile dysfunction (ED) is a multifactorial condition arising from vascular, neurological, hormonal, pharmacological, and psychological processes (Allen et al., 2023). Erectile response

is partly autonomically regulated: parasympathetic signalling facilitates penile blood flow, whereas sympathetic activity promotes detumescence (Chandler, 2015). ED may also generate anticipatory performance anxiety (Pyke, 2020). Within an allostatic framework, previous experience informs predictions about future demands and anticipatory physiological preparation (Sterling, 2012). Expectations of sexual failure may therefore contribute to physiological conditions that interfere with subsequent erectile response. Culturally informed beliefs about successful sex may further increase the stakes of these predictions.

Although EF is positively associated with sexual satisfaction, erectile difficulties do not necessarily preclude a positive sexual experience (Weiss & Brody, 2011). Men experiencing ED vary in their evaluations of their sexual lives, suggesting that erectile functioning and sexual satisfaction, although associated, are not interchangeable constructs (Albers et al., 2021; Althof et al., 2010). This variability is likely to reflect multiple factors. Sexual desire represents one such factor: poorer EF was associated with greater sexual bother particularly among men with high sexual desire (Bravi et al., 2020). Sexual-script flexibility may represent another potentially relevant process. When established sexual practices become difficult, couples may adapt their sexual scripts to accommodate changing circumstances. For instance, Bouchard et al. (2023) found that greater sexual-script flexibility in response to sexual challenges was cross-sectionally associated with greater sexual satisfaction among individuals and their partners. Thus, ED does not appear to determine sexual satisfaction uniformly, but its subjective consequences may depend partly on the meanings and sexual practices surrounding it.

Importantly, the availability of alternative sexual scripts and practices does not explain how couples come to regard them as satisfactory alternatives. Thomé

(2025) found that non-penetrative practices could remain positioned as “inferior scripts” even when penetration is impracticable. Adaptation may require partners to renegotiate what they understand satisfying sex to mean. Sexual communication provides one potential mechanism for this renegotiation. At the interpersonal level, sexual scripts are constructed between partners rather than simply reproduced from cultural expectations (Pham, 2016). Communication allows partners to express preferences and needs, negotiate activities, and revise shared expectations concerning sexual interaction. This process is particularly visible among queer couples, for whom heteronormative scripts may provide nonapplicable guidance (Sorg et al., 2026). This does not imply equivalence between queer experiences and heterosexual couples experiencing ED; rather, queer research illustrates a broader relational process.

Meta-analytic evidence links sexual communication with sexual satisfaction, with communication quality showing a stronger association than frequency or self-disclosure (Mallory, 2022). Communication quality — characterised by openness, comfort, and effectiveness (Sorg et al., 2026) — may be especially relevant to script disruption because adaptation requires effectively negotiating needs, expectations, and meanings. Sexual communication may also be relevant to physiological sexual response, as meta-analytic evidence associates better communication with sexual arousal and several domains of sexual function, including EF (Mallory et al., 2019). One possible mechanism may also involve performance anxiety reduction (Pyke, 2020), although psychophysiological interference remains to be established.

Male military veterans constitute a theoretically relevant population in which these processes may intersect. Erectile difficulties represent a meaningful

sexual-health concern among veterans and may coexist with physical injury, chronic health conditions, medication use, psychological distress, and PTSD-related symptoms, providing multiple potential biological and psychological pathways to impaired EF (Tran et al., 2015; Way et al., 2023). Sexual difficulties can affect veterans' self-perception, intimacy, and relationships, although their subjective significance is not uniform (Bird et al., 2021; Helmer et al., 2015). Military identities may additionally place particular emphasis on bodily capability and performance. Keeling and Sharratt (2023) describe how combat-related changes to the body can challenge veterans' masculinity, attractiveness, identity, and intimate relationships. For veterans to whom sexual functioning is personally significant, erectile difficulties may involve an intersection of physiological vulnerability and culturally informed expectations of masculine and bodily performance.

Research Question

Does perceived sexual communication quality moderate the association between erection hardness and sexual satisfaction among partnered male military veterans?

Rationale and Hypothesis

Emerging veteran rehabilitation research suggests that adaptation to sexual difficulties may involve shifts from performance-oriented sexual scripts centred on penetration towards pleasure-oriented understandings of sex (Ellis et al., 2021). However, the relational processes associated with such adaptation remain unclear. Research on military relationships suggests that communication, trust, and dyadic coping may support adjustment following deployment, trauma, and injury (Melvin et al., 2015; Sayers et al., 2018), although communication is not uniformly beneficial

and may depend on its quality and relational context (Bakhurst et al., 2018; Sayers et al., 2018). It is therefore unclear whether sexual communication quality changes the extent to which sexual satisfaction is associated with erection hardness among male military veterans.

It is hypothesised that sexual communication quality will moderate this association, such that lower erection hardness will be less strongly associated with lower sexual satisfaction at higher levels of perceived sexual communication quality.

Aim

To investigate the role of sexual communication in sexual wellbeing among partnered male military veterans.

Objectives

- To examine associations between erection hardness, perceived sexual communication quality, and sexual satisfaction.
- To test whether perceived sexual communication quality moderates the association between erection hardness and sexual satisfaction.
- To examine the association between erection hardness and sexual satisfaction at different levels of perceived sexual communication quality.

Research Design

The study will employ a quantitative cross-sectional moderation design, with erection hardness as the predictor, perceived sexual communication quality as the moderator, and sexual satisfaction as the outcome. Moderation will be interpreted statistically rather than causally. Veterans rather than veteran–partner dyads will be recruited as an initial individual-level test of the hypothesis.

Materials

Erection hardness will be assessed using the Erection Hardness Score (EHS; Mulhall et al., 2007), a validated single-item self-report measure (Appendix A). It was selected over broader measures such as the IIEF (Rosen et al., 1997) because it remains conceptually separable from sexual satisfaction. However, two response categories retain penetration-centred assumptions.

Quality of sexual communication will be measured using the six-item version of the Dyadic Sexual Communication Scale (DSC; Catania, 2013; Sorg et al., 2026). The measure assesses participants' perceived effectiveness, openness, and comfort in communicating with a primary partner about sexual matters (Appendix B).

Sexual satisfaction will be assessed using the Global Measure of Sexual Satisfaction (GMSEX; Lawrance & Byers, 1995). The measure provides a global evaluation of the participant's current sexual relationship without defining satisfaction through erection, penetration, or orgasm (Appendix C). It allows EF and subjective sexual satisfaction to remain distinct.

As the selected measures do not prescribe a fixed common recall period, the temporal frame used in the present study will be determined in consultation with the research supervisor to ensure consistency across measures and participants.

A brief demographic questionnaire will record age and relationship characteristics necessary to describe the sample (Appendix D). Additional service- or health-related information will not be collected given the sensitivity of military and health information.

Prior to recruitment, validated local-language versions of the measures will be used where available. Otherwise, the questionnaire will undergo translation and cultural adaptation before being pilot-tested with a small sample to assess its clarity, usability, and cultural fitness.

Participants

The proposed population is partnered male military veterans living in Armenia. Armenia provides a relevant post-conflict context, where recent hostilities have generated substantial psychological and rehabilitation needs among military personnel (Arakelyan et al., 2023; Azizian et al., 2023), while sexual wellbeing remains comparatively underexplored.

Recruitment could occur through Armenian veteran organisations and relevant online networks. Collaboration with Armenian researchers and veteran organisations would be important for culturally appropriate terminology and recruitment.

Eligibility will require participants to be aged 18 years or older, identify as male, have previous military service, and currently have an intimate partner with whom sexual communication is relevant. Participants will not be required to have diagnosed erectile dysfunction.

Data Analysis Plan

Data will be screened for missing values, influential observations, and relevant regression assumptions, followed by descriptive statistics and bivariate correlations. An a priori power analysis in G*Power 3.1 for the interaction effect (fixed-model multiple regression, R^2 increase; $f^2 = .05$, $\alpha = .05$, power = .80) indicated a required

sample of approximately 160 participants. The hypothesis will be tested using multiple regression, with erection hardness, perceived sexual communication quality, and their interaction entered as continuous predictors. A significant interaction will be probed using simple slopes at lower, mean, and higher levels of communication quality.

Procedure

Following ethical approval and pilot study, recruitment materials will invite veterans to participate in an anonymous online study examining EF, sexual satisfaction, and sexual communication within intimate relationships.

Before providing informed consent, participants will receive information explaining the sensitive nature of questions concerning sexual functioning, relationships, and health (Appendix E). Participants will then complete the questionnaire. The survey is expected to take approximately 10–15 minutes.

Participants may skip questions or withdraw before submission, after which they will receive a debrief with relevant support resources.

Ethics

The study concerns sensitive aspects of sexual functioning and intimate relationships within a population in which some individuals may also have experienced physical or psychological trauma. Ethical procedures will follow the general principles of respect for autonomy, privacy and dignity, scientific integrity, social responsibility outlined in the BPS Code of Human Research Ethics (British Psychological Society [BPS], 2021a). Particular attention is warranted because the BPS identifies research concerning sensitive sexual behaviours and participants with

experiences of trauma as requiring careful consideration of risk, consent, privacy, and potential stigmatisation. As data will be collected online, the BPS Ethics Guidelines for Internet-mediated Research (2021b) will additionally inform procedures concerning consent, anonymity, data security, and withdrawal.

Veteran status will not itself be treated as evidence of vulnerability or trauma. However, military and veteran research requires particular attention to duty, trust, confidentiality, and institutional influence, which may affect voluntariness and disclosure (Heward & Li, 2026; Townsend et al., 2026). Recruitment will therefore be opt-in; partner organisations may distribute study information but will not identify participants. Materials will explicitly describe the sexual nature of the study, emphasise voluntary participation, and avoid presenting erectile difficulties as an expected consequence of military service.

Data minimisation will be applied: no identifying information or detailed military or medical history will be collected, and anonymous responses will not be disclosed to military or veteran organisations, healthcare providers, or partners. Participants may skip questions or discontinue before submission. Locally appropriate sexual-health, relationship, and psychological-support resources will be provided before participation and in the debrief (Baillie Abidi et al., 2024).

Limitations

First, the cross-sectional design prevents causal inference. A significant interaction would indicate that the association between erection hardness and sexual satisfaction differs according to perceived sexual communication quality, but would not establish communication as a protective mechanism. Moreover, sexual satisfaction may itself influence communication, and both may be embedded within

broader relational processes, including intimacy and overall relationship quality (Beaulieu et al., 2023; Mallory, 2022).

Second, sexual communication is a dyadic process but is assessed only from the veteran's perspective, preventing analysis of how partners jointly negotiate changing sexual practices or meanings. Moreover, high-quality communication may facilitate negotiation without eliminating erectile difficulty as a meaningful loss (Fisher et al., 2009; O'Connor et al., 2012). Finally, the measure's emphasis on explicit discussion may not capture culturally variable or indirect forms of sexual communication (Mallory, 2022; Marlow et al., 2010).

Third, biological interpretation is limited because the EHS is a self-report measure rather than a direct physiological recording (Mulhall et al., 2007). Without autonomic measures, the study cannot determine whether communication relates to physiological threat responses or to the subjective meaning of erectile difficulty. EHS scores will therefore be interpreted as self-reported erection hardness rather than evidence of a specific biological mechanism or clinical ED.

Finally, findings may have limited generalisability because veterans are heterogeneous across cultural context, service history, disability, relationship and sexual identities, and time since service (McCaslin et al., 2021; Sachdev & Dixit, 2024). Future longitudinal and dyadic research incorporating psychophysiological measures could address these limitations.

Timeline

Task	Period
Final literature review; consultation with Armenian researchers and veteran organisations; refinement of eligibility criteria, terminology, and measures.	14.09.2026-2.10.2026
Ethics application; translation and cultural adaptation where required; pilot testing of survey materials.	5.10.2026-31.10.2026
Recruitment and data collection.	1.11.2026-31.12.2026
Data cleaning and preliminary analyses.	1.01.2027-31.01.2027
Moderation analysis and interpretation.	1.02.2027-28.02.2027
Writing, revision, and final preparation of the research report.	1.03.2027-31.03.2027

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Appendix A

Erection Hardness Score (EHS)

Question: How would you rate the hardness of your erection?

Score	Response
0	Penis does not enlarge
1	Penis is larger but not hard
2	Penis is hard but not hard enough for penetration
3	Penis is hard enough for penetration but not completely hard
4	Penis is completely hard and fully rigid

Source: Mulhall et al. (2007)

Appendix B

Dyadic Sexual Communication Scale — 6-item version

Questions:

1. I find some sexual matters are too upsetting to talk about with my primary partner.
2. I think it is difficult for my primary partner to tell me what he/she likes to do sexually.
3. It is easy for me to tell my primary partner what I do or don't like to do during sex.
4. My primary partner hardly ever talks to me when I want to talk about our sex life.
5. My primary partner really cares about what I think about sex.
6. Talking about sex with my primary partner is usually fun for the both of us.

Response scale:

1 = Completely disagree

2

3

4

5

6 = Completely agree

Scoring: Items 1, 2, and 4 are reverse-scored before all item scores are summed to produce a total score ranging from 6 to 36. Higher total scores indicate higher perceived quality of sexual communication.

Source: Source: Dyadic Sexual Communication Scale (Catania, 2013), six-item version as used by Sorg et al. (2026).

Appendix C

Global Measure of Sexual Satisfaction (GMSEX)

Question: In general, how would you describe your current sex life?

	7	6	5	4	3	2	1	
Very Good	☐	☐	☐	☐	☐	☐	☐	Very Bad
Very Pleasant	☐	☐	☐	☐	☐	☐	☐	Very Unpleasant
Very Positive	☐	☐	☐	☐	☐	☐	☐	Very Negative
Very Satisfying	☐	☐	☐	☐	☐	☐	☐	Very Unsatisfying
Very Valuable	☐	☐	☐	☐	☐	☐	☐	Worthless

Scoring: The five item scores are summed to produce a total score ranging from 5 to 35, with higher scores indicating greater sexual satisfaction.

Source: Lawrance and Byers (1995).

Appendix D

Demographic and Eligibility Questionnaire

1. What is your age?

____ years

2. How do you describe your gender?

☐ Man

☐ Another gender identity: _____

☐ Prefer not to say

3. Have you previously served in the military or armed forces?

☐ Yes

☐ No

☐ Prefer not to say

4. What best describes your current relationship situation?

☐ In a relationship with one partner

☐ In a relationship with more than one partner

☐ Married / civil partnership

☐ Another ongoing intimate relationship: _____

☐ Not currently in an intimate relationship

☐ Prefer not to say

5. Approximately how long have you been in your current relationship?

____ years ____ months

Appendix E

Participant Information Sheet

Date: 14.09.2026

Sponsoring institution: University of Essex

Principle Investigator: Dr Mark Ashton-Smith

Name of project

Sexual Communication as a Moderator of the Association Between Erection Hardness and Sexual Satisfaction Among Male Military Veterans

Purpose of the research

The purpose of this study is to investigate the relationship between erection, sexual communication, and sexual satisfaction among partnered male military veterans. In particular, the study examines whether the association between erection and sexual satisfaction differs depending on participants' perceived quality of sexual communication with their partner.

The study aims to contribute to a better understanding of sexual wellbeing among military veterans and of the relational factors that may be associated with satisfying sexual experiences across different levels of erectile functioning.

What is involved in participating?

If you choose to participate, you will be asked to complete an online questionnaire about your erection hardness, sexual satisfaction, and communication about sexual matters with your partner. You will also be asked to provide limited demographic information, such as your age and relationship characteristics.

The questionnaire is expected to take approximately 10–15 minutes to complete. Some questions concern personal and potentially sensitive aspects of sexual functioning and intimate relationships. You may choose not to answer any question that you do not wish to answer.

Benefits and risks of participating

There is no guaranteed direct personal benefit from taking part in this study. However, your participation may contribute to a better understanding of sexual wellbeing and intimate relationships among military veterans and may help inform future research and therapeutic practice in this area.

Some participants may experience discomfort, embarrassment, or emotional distress when answering questions about sexual functioning, sexual satisfaction, or intimate relationships. You are not required to answer any question that makes you uncomfortable and may stop participating at any time before submitting the questionnaire. Information about relevant sexual-health, relationship, and psychological-support services will be provided.

Procedures for withdrawal

Participation is entirely voluntary. You may stop completing the questionnaire at any point before submission without giving a reason and without any disadvantage to you.

Because the study does not collect identifying information, it may not be possible to identify and remove your responses after you have submitted the questionnaire at any stage.

Usage of the data

The information in this study will be used only for research purposes and in ways that will not reveal who you are. UK laws may require us to show information to university or government officials who are responsible for monitoring the safety of this study.

Your identifying information will not be shared with others outside this research study. Only the research team will have access to information that identifies you to carry out this research study. You will not be identified in any publication from this study. Your identifying information will be replaced with codes.

During the project, information from this study will be safeguarded such that only the research staff can access it. If your data is used for publication or other academic purposes any personal information that could identify you will be removed or changed before files are shared with other researchers or results are made public.

You can withdraw all or part of the data collected from you in the study before a month after the project completion date, after which the data may be used for publication or other academic purposes.

Any directly identifying elements in this research project (such as name and address or personal health information) will be stored separately from the research data disseminated for academic purposes. This information will be kept for the duration of the study plus an additional month and then will then be destroyed or de-identified, meaning we will replace your identifying information with a code that does not directly identify you.

Duration of project (start & completion dates): 1.11.2026-31.12.2026

How to file a complaint

If you wish to file a complaint about this research project, you can do so by contacting the Ethics Committee at the British Psychological Association ([link](#)).

Contact details for Principle Investigator

If you have any questions, would like additional information, or want to withdraw your data, you can contact the researcher/s directly.

Principle Investigator: Dr Mark Ashton-Smith, mark.ashton-smith@kaplan.com

Researcher: Anfisa Rekina, ar25534@essex.ac.uk

Please tick the appropriate boxes

Yes No

Taking part in the study

I have read and understood the Research Project Information Sheet or it has been read to me. I have been able to ask questions about the study and my questions have been answered to my satisfaction.

☐ ☐

I consent voluntarily to be a participant in this study and understand that I can withdraw from the study at any time, without having to give a reason.

☐ ☐

I understand that taking part in the study involves (relevant responsibilities of participant):

☐ ☐

Completing an online questionnaire lasting approximately 10–15 minutes. The questionnaire will include questions about erection, sexual satisfaction, sexual communication with my partner, and limited demographic and relationship information.

Risks associated with participating in the study

I understand that taking part in the study involves the following risks (omit if not relevant):

☐ ☐

Some questions concern sensitive aspects of sexual functioning and intimate relationships and may cause discomfort, embarrassment, or emotional distress. I understand that I may choose not to answer questions that make me uncomfortable and may discontinue the questionnaire before submission.

Use of the information in the study

I understand that information I provide will be used for (planned outlets such as reports, publications, websites, video channels)

☐ ☐

The information collected will be used for an MSc research project at the University of Essex Online. Findings may be presented in the research report and related academic presentations and may potentially contribute to future academic publications. Only aggregated and anonymised findings will be reported, and individual participants will not be identifiable in research outputs.

Please tick the appropriate boxes

Yes No

I understand I can withdraw all or part of the data collected from me in the study before a month after the project completion date, after which the findings may be used for publication or other academic purposes.

☐ ☐

I understand that personal information collected about me that can identify me, such as (e.g. my name or health details) will not be shared beyond the study team.

☐ ☐